9. LEPROSY

1. Disease Name

Leprosy

2. Programme

National Leprosy Eradication Programme (NLEP), Ministry of Health & Family Welfare, Government of India

3. Overview

- Leprosy is a chronic infectious disease.
- The disease primarily affects the skin and peripheral nerves.
- Leprosy can cause nerve damage, loss of sensation, muscle weakness and disability if diagnosis and treatment are delayed.
- Leprosy is curable with multidrug therapy (MDT).
- The National Leprosy Eradication Programme provides diagnosis and treatment services free of cost through public health facilities.
- The current revised classification and treatment protocol was implemented throughout India from **1 April 2025**.

4. Causative Organism

- **Mycobacterium leprae** is the causative organism of leprosy.
- The organism primarily affects peripheral nerves and the skin.

5. Transmission

- Transmission is associated mainly with prolonged exposure to an untreated person with leprosy.
- The respiratory tract is an important route involved in transmission.
- Respiratory droplets from an untreated infectious person can transmit the organism.
- The risk of transmission decreases substantially after effective treatment is started.
- The incubation period of leprosy is long and variable.

6. Main Signs and Symptoms

Skin-related signs

- Pale or hypopigmented skin patches.
- Reddish skin patches.
- Loss or reduction of sensation over a skin lesion.
- Thickened or infiltrated skin.
- Nodules may occur in some forms of disease.

Nerve-related signs

- Thickened peripheral nerves.
- Numbness.
- Loss of sensation.
- Muscle weakness.
- Nerve pain or tenderness.
- Weakness of muscles supplied by affected nerves.

Other possible manifestations

- Eye involvement.
- Nasal involvement.
- Ulcers caused by loss of sensation.
- Deformities caused by prolonged nerve damage.

7. Cardinal Signs Used for Diagnosis

A diagnosis of leprosy is supported by the presence of characteristic clinical or laboratory findings.

Cardinal Sign 1

- A pale or reddish skin lesion with definite loss of sensation.

Cardinal Sign 2

- Peripheral nerve involvement.
- Nerve involvement may include nerve thickening.
- Nerve involvement may be associated with loss of sensation.
- Nerve involvement may be associated with weakness of muscles supplied by the affected nerve.

Cardinal Sign 3

- Demonstration of **Mycobacterium leprae** in an appropriate laboratory specimen.

8. Revised Classification Under NLEP

Feature	Paucibacillary (PB)	Multibacillary (MB)
Skin lesions	1–5	More than 5
Nerve involvement	No nerve involvement	One or more nerves involved
Bacilli in slit-skin smear	Absent/negative	Present/positive

The revised classification emphasizes lesion count, nerve involvement and laboratory findings.

9. Diagnosis

Diagnosis is based primarily on clinical assessment.

Important diagnostic components include:

- Examination of skin lesions.
- Testing sensation over suspected skin lesions.
- Examination of peripheral nerves.
- Assessment of sensory function.
- Assessment of motor function.
- Slit-skin smear or other appropriate laboratory testing when indicated.

10. Treatment

- Leprosy is treated with **multidrug therapy (MDT)**.
- The current NLEP protocol uses a three-drug MDT regimen.
- The three drugs are:
    - Rifampicin
    - Dapsone
    - Clofazimine
- The revised protocol uses the three-drug regimen for both PB and MB cases.
- Current treatment duration under the revised NLEP protocol is:
    - PB: **6 months**

    - MB: **12 months**
- Treatment should be completed according to the prescribed regimen.
- Diagnosis and treatment services are provided free of cost through public health facilities.

11. Leprosy Reactions

Leprosy reactions are inflammatory episodes that may occur before, during or after treatment.

Type 1 Reaction

- Also called a reversal reaction.
- Existing skin lesions may become red, swollen, warm or tender.
- Nerve inflammation may occur.
- Nerve pain or tenderness may occur.
- Loss of nerve function can worsen during a reaction.

Type 2 Reaction

- Also called erythema nodosum leprosum (ENL).
- More commonly associated with multibacillary disease.
- May produce tender skin nodules.
- Fever may occur.
- Joint symptoms may occur.
- Nerve involvement may worsen.

12. Complications

Untreated or inadequately managed leprosy may result in:

- Permanent nerve damage.
- Loss of sensation.
- Muscle weakness.
- Paralysis.
- Hand deformity.
- Foot deformity.
- Ulcers.

- Eye complications.
- Permanent disability.

13. Disability Prevention

NLEP emphasizes:

- Early detection.
- Early initiation of treatment.
- Monitoring of nerve function.
- Prevention of disability.
- Medical rehabilitation.
- Management of complications.
- Reconstructive surgery for selected cases requiring specialized care.

14. Prevention and Control

Important programme strategies include:

- Early detection of new cases.
- Complete treatment.
- Household contact surveys.
- Contact screening.
- Post-exposure prophylaxis for eligible contacts.
- Health education.
- Community awareness.
- Reduction of stigma and discrimination.
- Surveillance and reporting.
- Disability prevention and rehabilitation.

15. When to Seek Medical Evaluation

Medical evaluation is recommended when a person develops:

- A persistent pale or reddish skin patch with reduced sensation.
- Numbness.
- Loss of sensation in the hands or feet.

- Unexplained weakness of the hands or feet.
- Thickened or painful peripheral nerves.
- Recurrent painless injuries or ulcers associated with loss of sensation.
- New deformity of the hands, feet or eyes.

16. Key Public-Health Message

- Leprosy is curable.
- Early diagnosis helps prevent nerve damage and disability.
- Treatment is available free of cost through public health facilities under NLEP.
- Persons affected by leprosy should not be stigmatized or discriminated against.

17. Quick Fact Sheet

Field	Information
Disease	Leprosy
Causative organism	Mycobacterium leprae
Main organs affected	Skin and peripheral nerves
Major clinical signs	Skin lesions with sensory loss and peripheral nerve involvement
Classification	Paucibacillary and Multibacillary
Current NLEP MDT	Rifampicin + Dapsone + Clofazimine
PB treatment duration	6 months
MB treatment duration	12 months
Major complication	Permanent nerve damage and disability
Prevention focus	Early detection, treatment, contact measures and disability prevention
National programme	National Leprosy Eradication Programme

Official Reference

Ministry of Health & Family Welfare, Government of India — Directorate General of Health Services, National Leprosy Eradication Programme (NLEP).

10. CHOLERA

1. Disease Name

Cholera

2. Source

National Centre for Disease Control (NCDC), Ministry of Health & Family Welfare, Government of India

3. Overview

- Cholera is an acute enteric infection.
- It is caused by **Vibrio cholerae**.
- Infection occurs through ingestion of contaminated food or water.
- Cholera is strongly associated with inadequate sanitation and unsafe water.
- Severe cholera can cause sudden, profuse watery diarrhoea.
- Severe fluid loss can lead to dehydration and death.
- Outbreaks can spread rapidly because of the short incubation period.
- With appropriate and timely treatment, cholera can be successfully managed.

4. Causative Organism

- **Vibrio cholerae** is the causative bacterium.
- Infection occurs when the organism is ingested through contaminated food or water.

5. Transmission

- Cholera is transmitted mainly through contaminated water.
- Cholera can also be transmitted through contaminated food.
- Faecal contamination of drinking-water sources is an important route of transmission.
- Poor sanitation increases transmission.
- Unsafe food preparation and handling can contribute to spread.
- Flooding can increase the risk of contamination of drinking-water sources.
- Population displacement and overcrowding can increase outbreak risk.

6. Incubation Period

- The incubation period can be very short.
- NCDC material describes an incubation period ranging from approximately **2 hours to 5 days**.
- The short incubation period can contribute to rapid outbreak expansion.

7. Symptoms

Main symptom

- Acute watery diarrhoea.

Characteristic severe presentation

- Profuse watery diarrhoea.
- **Rice-water stools** may occur.
- Vomiting may occur.
- Rapid loss of body fluids may occur.

Severe disease

- Severe dehydration.
- Weakness.
- Circulatory compromise.
- Death can occur rapidly if severe dehydration is not treated.

NCDC identifies profuse diarrhoea with rice-water stools as a clinical pattern that should raise suspicion of cholera.

8. Asymptomatic Infection

- A substantial proportion of people infected with cholera may not develop symptoms.
- Asymptomatic infected individuals can still contribute to environmental contamination through shedding of the organism in faeces.

9. Clinical Suspicion

Cholera should be considered when a person has:

- Profuse acute watery diarrhoea.
- Rice-water stools.

- Vomiting associated with severe diarrhoea.
- Rapid development of dehydration.
- A compatible illness during a known or suspected outbreak.
- A history of exposure to potentially contaminated water or food.

10. Major Complication

The major immediate complication of severe cholera is:

Rapid dehydration due to large-volume fluid loss.

Severe dehydration can progress to:

- Circulatory failure.
- Shock.
- Organ dysfunction.
- Death.

11. Treatment

Rehydration

- **Rehydration is essential.**
- Oral rehydration is used when the patient can drink adequately.
- Intravenous fluid replacement is required when dehydration is severe or oral replacement is inadequate.

Antibiotic therapy

- Antibiotics are used as **adjunctive therapy** in appropriate cases.
- NCDC's National Treatment Guidelines list doxycycline as a first-line option for adults in the cited treatment guidance.
- Azithromycin is listed for children and pregnant women in the cited guideline.
- Alternative antibiotic options are also provided in the NCDC guideline.
- Antibiotic selection should follow current clinical guidance and patient-specific considerations.

Treatment principle

Replace fluid and electrolytes first and rapidly.

12. Outbreak Management

NCDC identifies the following important outbreak-control measures:

- Epidemiological investigation.
- Establishment of cholera treatment centres/units where required.
- Appropriate isolation facilities for seriously ill patients.
- Provision of safe drinking water.
- Provision of safe food.
- Improvement of sanitation.
- Proper disposal of human waste and sewage.
- Appropriate disinfection.
- Health education.
- Surveillance and monitoring.

13. Prevention

Safe Water

- Drink water from a safe source.
- Disinfect drinking water when contamination is suspected.
- During flood situations, appropriate water disinfection and chlorination are important.

Food Safety

- Cook food thoroughly.
- Reheat food thoroughly when required.
- Eat freshly prepared hot food where possible.
- Boil milk before drinking.
- Avoid uncooked food from unsafe sources.
- Avoid ice from unreliable sources.

Hand Hygiene

- Wash hands after contact with faecal matter.
- Wash hands before preparing food.
- Wash hands before eating.

Sanitation

- Dispose of human excreta safely.
- Prevent faecal contamination of drinking-water sources.
- Maintain proper sanitation facilities.

14. Vaccination

- Cholera vaccines may be used as an additional public-health measure in appropriate settings.
- Vaccination should not replace safe water, sanitation, hygiene and appropriate outbreak-control measures.
- Mass chemoprophylaxis is not recommended as a general strategy for controlling cholera outbreaks according to the NCDC cholera guidance.

15. High-Risk Situations

Risk increases with:

- Unsafe drinking water.
- Poor sanitation.
- Flooding.
- Population displacement.
- Crowding.
- Unsafe food handling.
- Disruption of water and sanitation infrastructure.

16. When to Seek Immediate Medical Care

Immediate medical assessment is required for:

- Profuse watery diarrhoea.
- Repeated vomiting.
- Signs of dehydration.
- Severe weakness.
- Dizziness or collapse.
- Reduced urine output.
- Rapid worsening of diarrhoea.

17. Key Public-Health Message

- Cholera is preventable and treatable.
- Rapid rehydration is the most important component of management.
- Safe drinking water, sanitation and hygiene are central to prevention.
- Suspected severe cholera should be treated urgently.

18. Quick Fact Sheet

Field	Information
Disease	Cholera
Causative organism	Vibrio cholerae
Main transmission	Contaminated food or water
Main symptom	Acute watery diarrhoea
Characteristic severe stool	Rice-water stool
Major complication	Severe dehydration
Incubation	Approximately 2 hours–5 days
Main treatment	Rapid rehydration
Severe dehydration	Intravenous fluid replacement
Antibiotics	Adjunctive treatment in appropriate cases
Prevention	Safe water, sanitation, hygiene and food safety
Outbreak control	Surveillance, treatment centres, safe water, sanitation and health education

Official References

National Centre for Disease Control (NCDC), Ministry of Health & Family Welfare, Government of India — National Treatment Guidelines for Antimicrobial Use in Infectious Diseases.

NCDC — “Cholera: A Continuing Challenge to Public Health.”

11. JAPANESE ENCEPHALITIS

1. Disease Name

Japanese Encephalitis (JE)

2. Source

National Centre for Disease Control (NCDC), Ministry of Health & Family Welfare, Government of India

3. Overview

- Japanese encephalitis is a mosquito-borne viral disease.
- It is a zoonotic disease.
- Japanese encephalitis can cause inflammation of the brain.
- Severe disease can cause neurological complications and death.
- Japanese encephalitis is one of the causes of **Acute Encephalitis Syndrome (AES)**.
- NCDC maintains technical material covering diagnosis, treatment, surveillance and prevention of JE/AES.

4. Causative Agent

- **Japanese encephalitis virus (JEV)** is the causative virus.
- JEV is an arbovirus.
- The virus is maintained in an animal–mosquito transmission cycle.

5. Vector and Transmission

- Japanese encephalitis is transmitted by infected mosquitoes.
- **Culex mosquitoes** are important vectors.
- The transmission cycle involves mosquitoes and animal hosts.
- Pigs and certain water birds are important amplifying/reservoir hosts in the natural cycle.
- Humans can become infected after being bitten by an infected mosquito.
- Human-to-human transmission through ordinary contact does not occur.

Transmission Sequence

Animal reservoir/amplifying host → infected mosquito → mosquito bite → human infection

6. Epidemiological Pattern

- JE transmission is associated with mosquito abundance.
- Transmission may increase during and after the monsoon period in affected areas.
- Environmental factors such as waterlogging and paddy cultivation can influence mosquito breeding.
- Presence of pigs and migratory/water birds can be relevant to transmission ecology.

7. Clinical Presentation

Japanese encephalitis may present with:

- Fever.
- Headache.
- Altered mental status.
- Neurological symptoms.
- Encephalitis.
- Seizures.
- Coma in severe disease.

Not every person infected with JEV develops encephalitis.

8. Acute Encephalitis Syndrome

AES is a clinical syndrome rather than a single disease.

- Japanese encephalitis is one of the important causes of AES.
- Other infectious and non-infectious causes can also produce AES.
- A patient with suspected encephalitis requires urgent clinical evaluation.
- Laboratory investigation is important to identify the cause.

9. Laboratory Diagnosis

Serological testing

- Detection of **JEV-specific IgM antibodies** by IgM-capture ELISA is an important diagnostic method.
- JEV-specific IgM can be detected in cerebrospinal fluid (CSF) or serum.
- Detection of JEV-specific IgM in CSF provides strong evidence of JE infection.
- Serum results require careful interpretation in areas with high JE vaccination coverage because antibodies may reflect previous infection or vaccination.

Molecular testing

- Viral RNA or genomic sequences may be detected in appropriate clinical specimens.
- Molecular testing can also be used for characterization of circulating virus genotypes.

Virus isolation

- Virus isolation can be performed using appropriate clinical or mosquito specimens in specialized laboratories.
- Laboratory confirmation should be performed using appropriate validated methods.

10. Treatment

- **There is no specific antiviral treatment for Japanese encephalitis.**
- Treatment is primarily supportive.
- Early referral to a hospital is important.
- Treatment should not be delayed while waiting for serological laboratory results in a clinically suspected encephalitis case.

Supportive Management

Important management principles include:

- Maintaining fluid and electrolyte balance.
- Maintaining a clear airway.
- Providing oxygen when clinically required.
- Managing convulsions appropriately.
- Providing adequate nursing care.
- Monitoring neurological status.

- Providing hospital-level care for severe cases.

11. Severe Disease

Severe JE may lead to:

- Severe encephalitis.
- Seizures.
- Altered consciousness.
- Coma.
- Neurological impairment.
- Death.

12. Prevention

Vector Control

NCDC identifies vector control as an important component of JE prevention.

Measures include:

- Reduction of mosquito breeding.
- Vector surveillance.
- Appropriate insecticide-based vector-control measures.
- Environmental management.
- Reduction of mosquito–human contact.

Vaccination

- Vaccination is an important component of the national JE prevention and control programme.
- Vaccination strategies are implemented according to national programme recommendations and affected-area risk.

13. Surveillance

JE/AES surveillance includes:

- Detection of suspected AES cases.
- Laboratory investigation.
- Confirmation of JE cases.
- Monitoring of cases and deaths.

- Monitoring of vaccination status.
- Vector surveillance.
- Assessment of environmental risk factors.
- Monitoring of geographical and seasonal patterns.

14. Risk Factors and Environmental Factors

Factors associated with JE transmission include:

- Increased mosquito density.
- Waterlogged areas.
- Paddy fields.
- Presence of pig farms.
- Presence of migratory birds.
- Seasonal increases in mosquito populations.
- Increased mosquito–human contact.

15. When to Seek Urgent Medical Care

Urgent medical care is required when fever is associated with:

- Altered consciousness.
- Confusion.
- Severe headache.
- Seizures.
- Abnormal behaviour.
- Neurological weakness.
- Neck stiffness.
- Coma.
- Rapid neurological deterioration.

16. Key Public-Health Message

- Japanese encephalitis is a serious mosquito-borne disease.
- Severe disease can cause encephalitis and neurological complications.
- Suspected encephalitis requires early hospital referral.

- There is no specific antiviral treatment for JE.
- Prevention depends on vaccination, surveillance and mosquito-control measures.

17. Quick Fact Sheet

Field	Information
Disease	Japanese Encephalitis
Causative agent	Japanese encephalitis virus
Disease type	Mosquito-borne viral disease
Important vector	Culex mosquitoes
Main clinical syndrome	Encephalitis
Relationship to AES	JE is one cause of AES
Main symptoms	Fever, headache and neurological manifestations
Severe manifestations	Seizures, altered consciousness, coma and neurological complications
Specific antiviral treatment	None
Main treatment	Supportive care
Important diagnosis	JEV-specific IgM by IgM-capture ELISA
Prevention	Vaccination and vector control
Surveillance	AES/JE case surveillance and laboratory confirmation

Official References

National Centre for Disease Control (NCDC), Ministry of Health & Family Welfare, Government of India — Manual of Zoonotic Diseases of Public Health Importance, Chapter: Japanese Encephalitis.

NCDC/NCVBDC — Guidelines on Clinical Management of Japanese Encephalitis and Revised Treatment Guidelines of AES including JE.

12. KALA-AZAR / VISCERAL LEISHMANIASIS

1. Disease Name

Kala-azar

2. Medical Name

Visceral Leishmaniasis (VL)

3. Source

Ministry of Health & Family Welfare, Government of India — National Center for Vector Borne Diseases Control (NCVBDC), Kala-azar Elimination Programme

4. Overview

- Kala-azar is also called visceral leishmaniasis.
- It is an infectious disease caused by **Leishmania parasites**.
- The disease is transmitted through the bite of an infected sandfly.
- Kala-azar primarily affects the internal organs and reticuloendothelial system.
- The disease commonly presents with prolonged fever, splenomegaly and weight loss.
- Untreated kala-azar can be fatal.
- Kala-azar is a major target of the national elimination programme.

5. Causative Agent

- Kala-azar is caused by **Leishmania parasites**.
- The parasite responsible for visceral leishmaniasis belongs to the *Leishmania donovani* complex in the Indian subcontinent.

6. Vector

- Kala-azar is transmitted by infected **female sandflies**.
- Sandflies acquire the parasite while feeding on an infected host.
- An infected sandfly can transmit the parasite during a subsequent blood meal.

Transmission Sequence

Infected host → sandfly acquires parasite → parasite develops in sandfly → infected sandfly bites human → human infection

7. Main Symptoms

The main clinical features include:

- Prolonged fever.
- Irregular fever.
- Weight loss.
- Enlarged spleen.
- Enlarged liver.
- Weakness.
- Anaemia.
- Progressive deterioration in untreated disease.

NCVBDC programme material identifies prolonged fever, splenomegaly and weight loss as the major clinical features of kala-azar.

8. Suspected Case

A kala-azar suspect is generally identified when a person from an endemic area has:

- Fever for more than two weeks.
- Enlarged spleen.
- Relevant epidemiological exposure.
- Lack of response to an appropriate course of antimalarial treatment when malaria has been considered.

The national programme uses a standardized clinical case definition before diagnostic testing.

9. Diagnosis

Rapid Diagnostic Test

- A quality-assured **rK39-based rapid diagnostic test (RDT)** is used in the programme for appropriate suspected cases.
- The test detects antibodies associated with visceral leishmaniasis.
- The rK39 test is used as an important field-level diagnostic tool.

Parasitological Confirmation

Parasitological confirmation can involve detection of the parasite in appropriate specimens.

Possible specimens include:

- Bone marrow aspirate.
- Splenic aspirate.
- Other appropriate tissue/material according to clinical and laboratory facilities.
- Parasitological testing should be performed by appropriately trained personnel at suitable facilities.

10. Important Diagnostic Consideration

- rK39-based testing should be used within the appropriate clinical case definition.
- A positive antibody test should be interpreted in the appropriate clinical and epidemiological context.
- Special situations such as relapse, reinfection and HIV–VL co-infection require appropriate specialist evaluation.
- Patients with high clinical suspicion and a negative rapid test may require further diagnostic evaluation.

11. Treatment

- Kala-azar requires **specific anti-leishmanial treatment**.
- The national programme has introduced treatment regimens based on effective anti-leishmanial medicines.
- The NCVBDC operational guidelines identify **liposomal amphotericin B** as an important treatment option.
- Other treatment options described in programme guidelines include:
    - Miltefosine.
    - Amphotericin B.
    - Combination treatment with appropriate anti-leishmanial medicines.
- Treatment selection depends on the national programme protocol, clinical condition and relevant patient factors.

12. Treatment Follow-up

Treatment outcomes should be monitored.

The kala-azar elimination programme distinguishes treatment outcomes based on:

- Initial outcome at completion of drug treatment.
- Final outcome during subsequent follow-up.

Patients should remain under appropriate follow-up after treatment to identify treatment failure or relapse.

13. Post-Kala-Azar Dermal Leishmaniasis (PKDL)

- PKDL is a dermal manifestation that can occur after visceral leishmaniasis.
- It produces skin manifestations.
- PKDL may resemble other skin diseases.
- PKDL can sometimes resemble leprosy.
- Identification and treatment of PKDL are important for the elimination programme because affected individuals can contribute to parasite transmission.
- NCVBDC provides specific guidelines for PKDL diagnosis and treatment.

14. Complications

Untreated or inadequately treated kala-azar may lead to:

- Progressive weakness.
- Severe wasting and weight loss.
- Anaemia.
- Enlargement of the spleen and liver.
- Severe systemic illness.
- Death.

Kala-azar is considered potentially fatal when untreated.

15. Prevention and Control

Early Case Detection

- Active case search is used in programme areas.
- Passive case detection occurs through health facilities.
- Suspected cases are evaluated and tested promptly.

- Early diagnosis reduces the period during which untreated infection can contribute to transmission.

Treatment

- Confirmed cases should receive appropriate complete treatment.
- Treatment outcomes should be monitored.

Vector Control

Integrated vector management is an important component of the elimination programme.

Measures include:

- Indoor residual spraying where indicated.
- Appropriate insecticide use.
- Vector surveillance.
- Reduction of human–sandfly contact.
- Environmental measures where appropriate.
- Monitoring of vector density and transmission risk.

16. Surveillance

Kala-azar surveillance includes:

- Passive case detection.
- Active case search.
- Reporting of kala-azar cases.
- Reporting of PKDL cases.
- Vector surveillance.
- Monitoring of disease trends.
- Mapping of affected areas.
- Use of surveillance information for planning vector-control activities.

17. Risk Factors

Risk of kala-azar is associated with:

- Residence in endemic areas.
- Exposure to infected sandflies.

- Conditions that increase human–vector contact.
- Poverty and poor living conditions.
- Environmental conditions favourable to sandfly transmission.

18. When to Seek Medical Evaluation

Medical evaluation is recommended for a person from an endemic area who develops:

- Fever lasting more than two weeks.
- Enlarged spleen.
- Unexplained weight loss.
- Persistent weakness.
- Anaemia or pallor.
- Persistent fever that does not respond appropriately to treatment for other suspected causes.

19. Key Public-Health Message

- Kala-azar is a serious but treatable disease.
- Early diagnosis and complete treatment are essential.
- Sandfly control is an important part of prevention.
- Active and passive surveillance support elimination.
- PKDL detection and management are important components of the elimination programme.

20. Quick Fact Sheet

Field	**Information**
Disease	Kala-azar
Medical name	Visceral leishmaniasis
Causative agent	Leishmania parasite
Vector	Infected female sandfly
Main symptoms	Prolonged fever, splenomegaly, weight loss
Other manifestations	Hepatomegaly, anaemia, weakness

Field	Information
Main diagnostic tool	rK39-based rapid diagnostic test in appropriate suspected cases
Additional diagnosis	Parasitological examination
Main treatment principle	Specific anti-leishmanial treatment
Important medicines in programme guidelines	Liposomal amphotericin B, miltefosine and amphotericin B
Major complication	Severe systemic disease and death if untreated
Related condition	Post-kala-azar dermal leishmaniasis (PKDL)
Prevention	Early diagnosis, complete treatment and vector control
Vector-control approach	Integrated vector management and indoor residual spraying where indicated
Surveillance	Active search, passive detection, disease surveillance and vector surveillance
National programme	Kala-azar Elimination Programme

Official References

National Center for Vector Borne Diseases Control (NCVBDC), Ministry of Health & Family Welfare, Government of India — Operational Guidelines on Kala-Azar (Visceral Leishmaniasis) Elimination in India, 2015.

NCVBDC — Accelerated Plan for Kala-azar Elimination.

NCVBDC — Kala-azar Elimination Programme guidelines and operational resources.